

Problem

Find the Laplace transform $F(s)$, given that $f(t)$ is:

(a) $2tu(t-4)$

(b) $5 \cos(t) \delta(t-2)$

(c) $e^{-t}u(t-t)$

(d) $\sin(2t)u(t-\tau)$

Step-by-step solution

Step 1 of 5

(a)

Consider the signal, $f(t) = 2tu(t-4)$.

Write signal $f(t)$ as follows:

$$\begin{aligned} f(t) &= 2tu(t-4) \\ &= 2(t-4+4)u(t-4) \\ &= 2(t-4)u(t-4) + 8u(t-4) \end{aligned}$$

The Laplace transform pairs:

$$\begin{aligned} f(t-a)u(t-a) &\xrightarrow{\mathcal{L}} e^{-as}F(s) \\ u(t-a) &\xrightarrow{\mathcal{L}} \frac{e^{-as}}{s} \end{aligned}$$

Apply the Laplace transform pairs by substituting 4 in place of a to find $F(s)$.

$$\begin{aligned} F(s) &= 2 \left(e^{-4s} \left(\frac{1}{s^2} \right) \right) + 8 \left(\frac{e^{-4s}}{s} \right) \\ &= \frac{2e^{-4s}}{s^2} + \frac{8e^{-4s}}{s} \\ &= \left(\frac{2}{s^2} + \frac{8}{s} \right) e^{-4s} \end{aligned}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\left(\frac{2}{s^2} + \frac{8}{s} \right) e^{-4s}}$.

Step 2 of 5

(b)

Consider the signal, $f(t) = 5 \cos(t) \delta(t-2)$.

Calculate the Laplace transform of the signal $f(t)$.

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_{0^-}^{\infty} f(t) e^{-st} dt \\ &= \int_{0^-}^{\infty} (5 \cos(t) \delta(t-2)) e^{-st} dt \\ &= 5 \cos(t) e^{-st} \Big|_{t=2} \\ &= 5 \cos(2) e^{-2s} \end{aligned}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{5 \cos(2) e^{-2s}}$.

Step 3 of 5

(c)

Consider the signal, $f(t) = e^{-t} u(t-\tau)$.

Write signal $f(t)$ as follows:

$$\begin{aligned} f(t) &= e^{-t} u(t-\tau) \\ &= e^{-(t-\tau+\tau)} u(t-\tau) \\ &= e^{-\tau} (e^{-(t-\tau)} u(t-\tau)) \end{aligned}$$

Write the Laplace transform pairs that are used to find $F(s)$.

$$\begin{aligned} e^{-t} u(t) &\xleftrightarrow{\mathcal{L}} \frac{1}{s+1} \\ f(t-a) u(t-a) &\xleftrightarrow{\mathcal{L}} e^{-as} F(s) \end{aligned}$$

Apply the Laplace transform pairs by substituting τ in place of a to find $F(s)$.

$$\begin{aligned} F(s) &= e^{-(\tau)} \left(e^{-\tau s} \left(\frac{1}{s+1} \right) \right) \\ &= \frac{1}{s+1} e^{-(s+1)\tau} \end{aligned}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\frac{1}{s+1} e^{-(s+1)\tau}}$.

Step 4 of 5

(d)

Consider the signal, $x(t) = \sin(2t)u(t - \tau)$.

Write signal $f(t)$ as follows:

$$\begin{aligned}x(t) &= \sin(2t)u(t - \tau) \\&= \sin(2(t - \tau + \tau))u(t - \tau) \\&= \sin(2[(t - \tau) + \tau])u(t - \tau) \\&= \sin(2(t - \tau))\cos(2\tau) + \cos(2(t - \tau))\sin(2\tau)\end{aligned}$$

Write the Laplace transform pairs that are used to find $X(s)$.

$$\cos at u(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2}$$

$$\sin at u(t) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2}$$

$$f(t - a)u(t - a) \xleftrightarrow{\mathcal{L}} e^{-as}F(s)$$

Step 5 of 5

Apply the Laplace transform pairs to find $X(s)$.

$$X(s) = X(s) = \left\{ \cos(2\tau)e^{-\tau s} \left(\frac{2}{s^2 + 2^2} \right) + \sin(2\tau)e^{-\tau s} \left(\frac{s}{s^2 + 2^2} \right) \right\}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{e^{-\tau s} \cos(2\tau) \left(\frac{2}{s^2 + 2^2} \right) + e^{-\tau s} \sin(2\tau) \left(\frac{s}{s^2 + 2^2} \right)}$.